

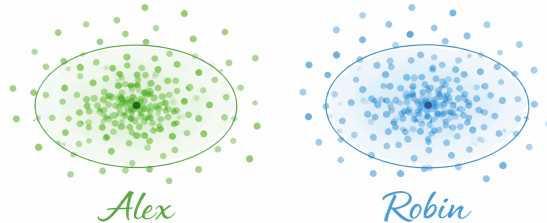
INTRODUCTION TO MACHINE LEARNING

PROJECT WEEK 3

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University of Luxembourg - Summer Semester
2025-2026

March 2026

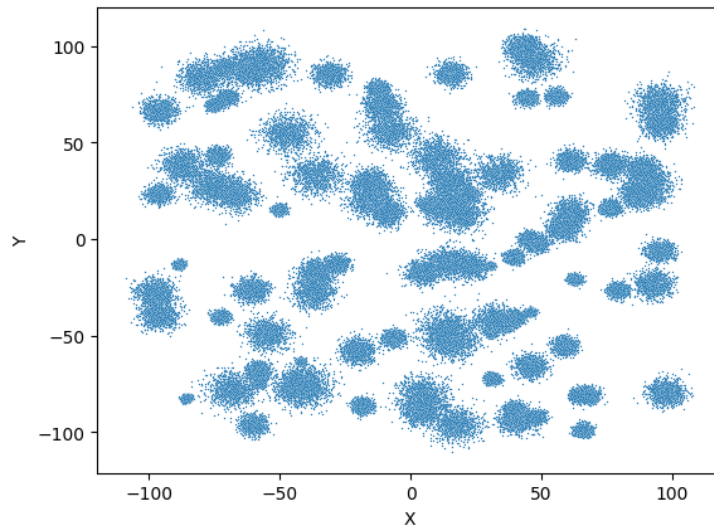


THE BIRCH CLASSIFICATION PROBLEM

UNDERSTANDING THE DATA

We observe that:

- ~ 30 -50 clusters



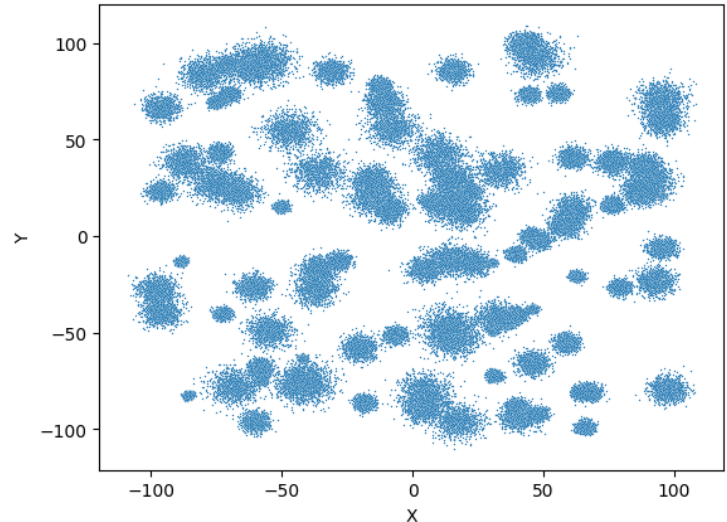
Raw data set (point size 1)

THE BIRCH CLASSIFICATION PROBLEM

UNDERSTANDING THE DATA

We observe that:

- ▶ ~ 30 -50 clusters
- ▶ Well-separated



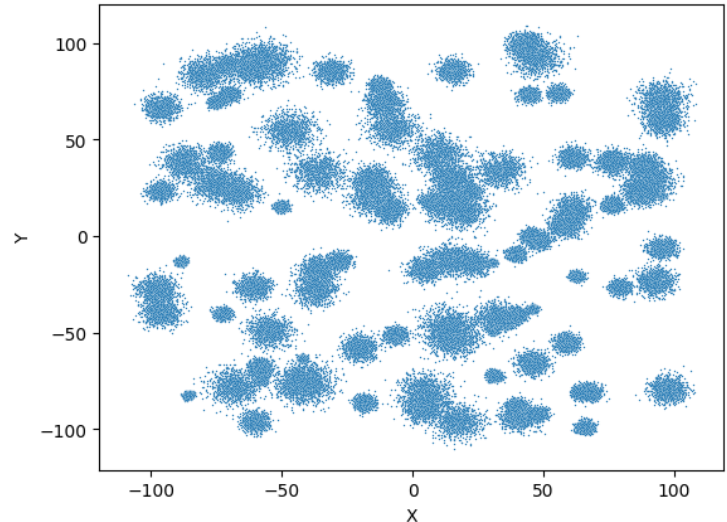
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THE BIRCH CLASSIFICATION PROBLEM

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We observe that:

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- ▶ Well-separated
- ▶ Similar densities, compact



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THE BIRCH CLASSIFICATION PROBLEM

UNDERSTANDING THE DATA

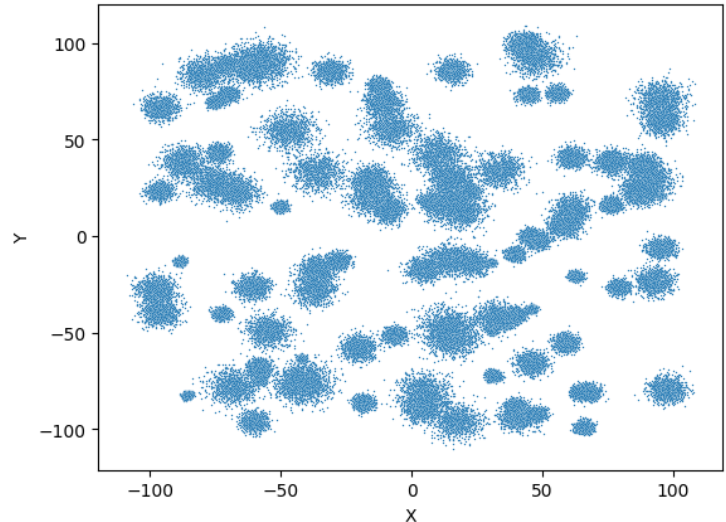
We observe that:

- ▶ ~ 30 -50 clusters
- ▶ Well-separated
- ▶ Similar densities, compact

Which would you choose?

- ▶ K-Means
- ▶ DBSCAN

Why?



Raw data set (point size 1)

THE BIRCH CLASSIFICATION PROBLEM

UNDERSTANDING THE DATA

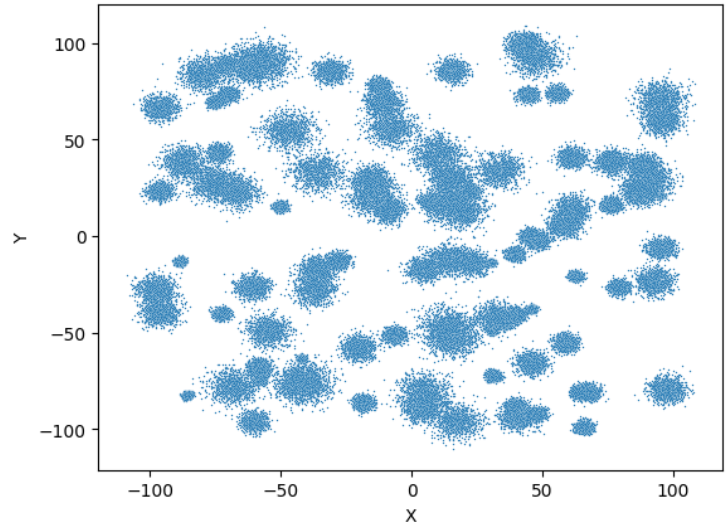
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- ▶ Similar densities, compact

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- ▶ K-Means
- ▶ **DBSCAN**

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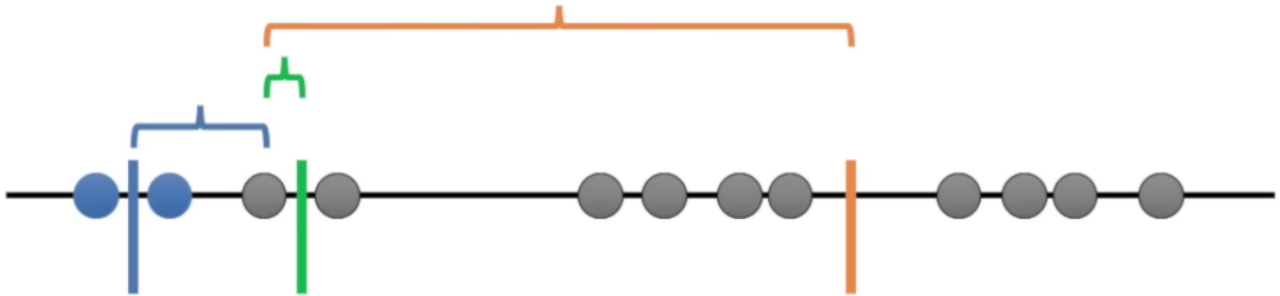


Raw data set (point size 1)

THE BIRCH CLASSIFICATION PROBLEM

WHY NOT K-MEANS?

- Clusters all spherical and about same densities, **but** clusters have different size



K-Means assigns a point to the cluster with closest distance

Source: <https://www.youtube.com/watch?v=4b5d3muPQmA>, as of 02.03.2026

THE BIRCH CLASSIFICATION PROBLEM

WHY NOT K-MEANS?

- ▶ Clusters all spherical and about same densities, **but** clusters have different size
- ▶ Hard to choose right K for amount of clusters

THE BIRCH CLASSIFICATION PROBLEM

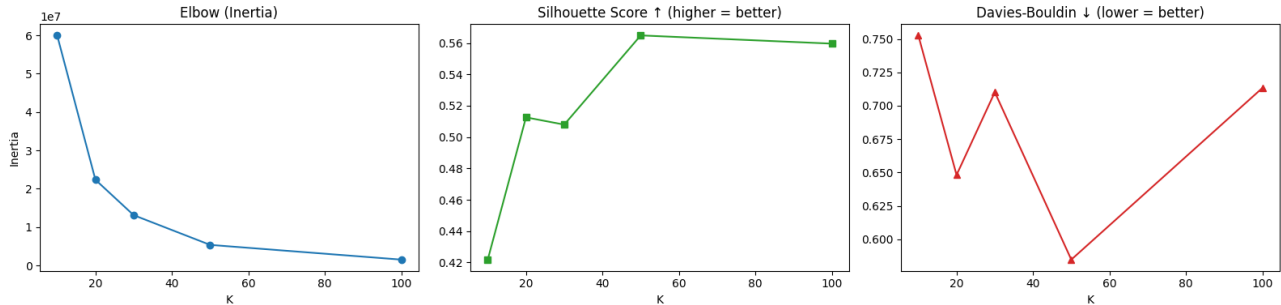
WHY NOT K-MEANS?

- ▶ Clusters all spherical and about same densities, **but** clusters have different size
- ▶ Hard to choose right K for amount of clusters
- ▶ Well, to choose K you can use metrics...

THE BIRCH CLASSIFICATION PROBLEM

WHY NOT K-MEANS?

- ▶ Clusters all spherical and about same densities, **but** clusters have different size
- ▶ Hard to choose right K for amount of clusters
- ▶ Well, to choose K you can use metrics...
- ▶ Below the best at K=50, but...

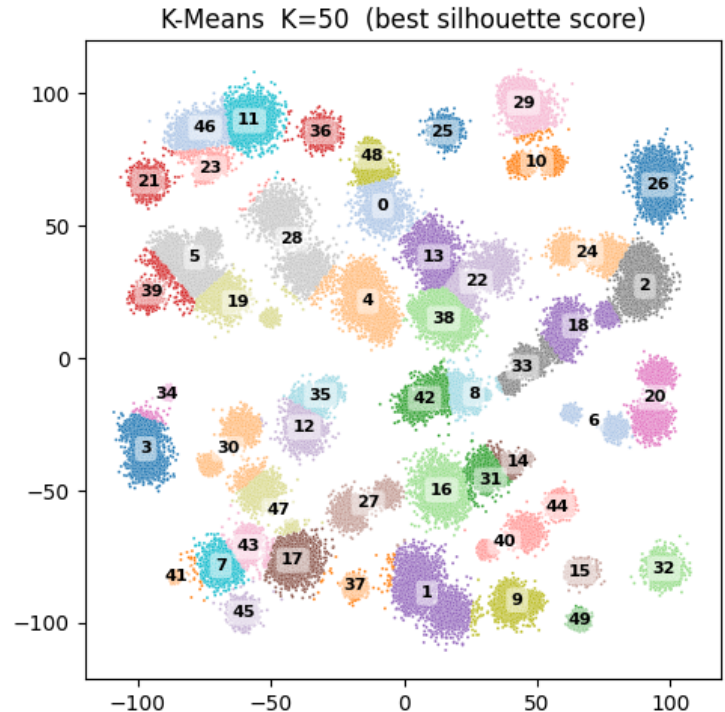


Metrics to determine best K for K-Means

THE BIRCH CLASSIFICATION PROBLEM

WHY NOT K-MEANS?

- Our best choice: $K=50$
- But does not solve problem that clusters have different sizes!



THE BIRCH CLASSIFICATION PROBLEM

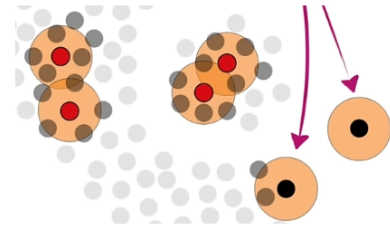
DBSCAN

- DBSCAN: Groups points that are densely packed (see image)

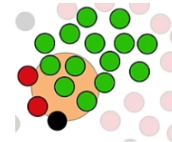
Image source:

<https://www.youtube.com/watch?v=RDZUdRSDook>, as of 02.03.2026

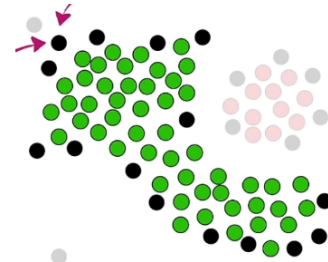
Core points



Assign neighboring core points to cluster



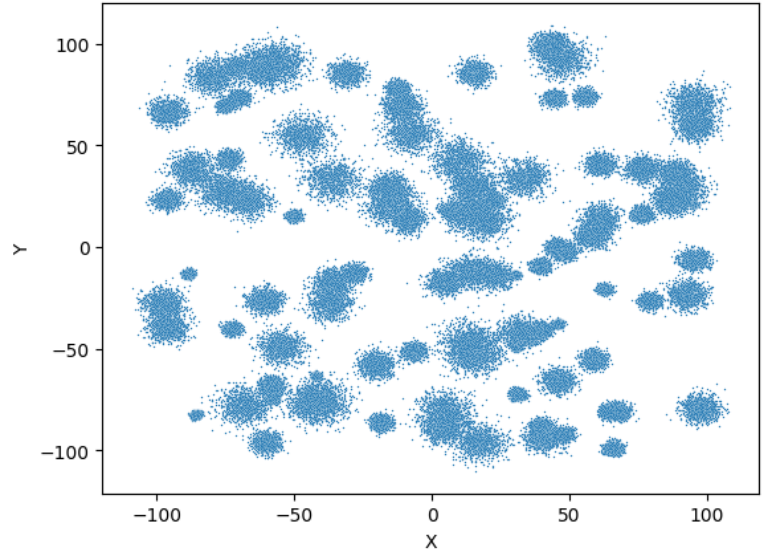
Add all non-core points within range to cluster



THE BIRCH CLASSIFICATION PROBLEM

DBSCAN

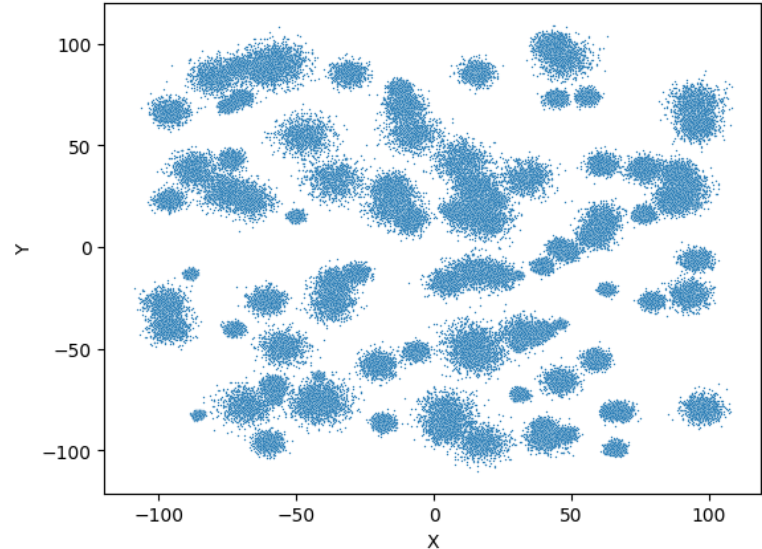
- ▶ DBSCAN: Groups points that are densely packed
- ▶ Clusters in data set are **dense**, but clusters not of equal size



THE BIRCH CLASSIFICATION PROBLEM

DBSCAN

- ▶ DBSCAN: Groups points that are densely packed
- ▶ Clusters in data set are **dense**, but clusters not of equal size
- ▶ Intuitively: Small circle ("eps") for core points



THE BIRCH CLASSIFICATION PROBLEM

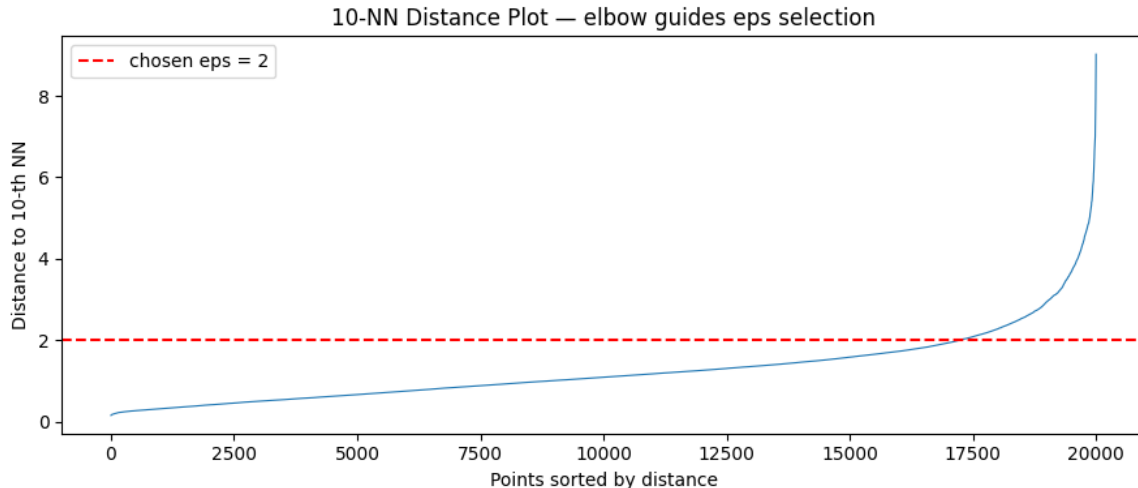
DBSCAN

- ▶ Intuitively, because dense clusters: Small circle ("eps") for core points
- ▶ Better: Confirm this by metrics

THE BIRCH CLASSIFICATION PROBLEM

DBSCAN

- ▶ Intuitively, because dense clusters: Small circle ("eps") for core points
- ▶ Better: Confirm this by metrics
- ▶ Sort distance of each point to the 10th closest neighbor (sample: 20k)

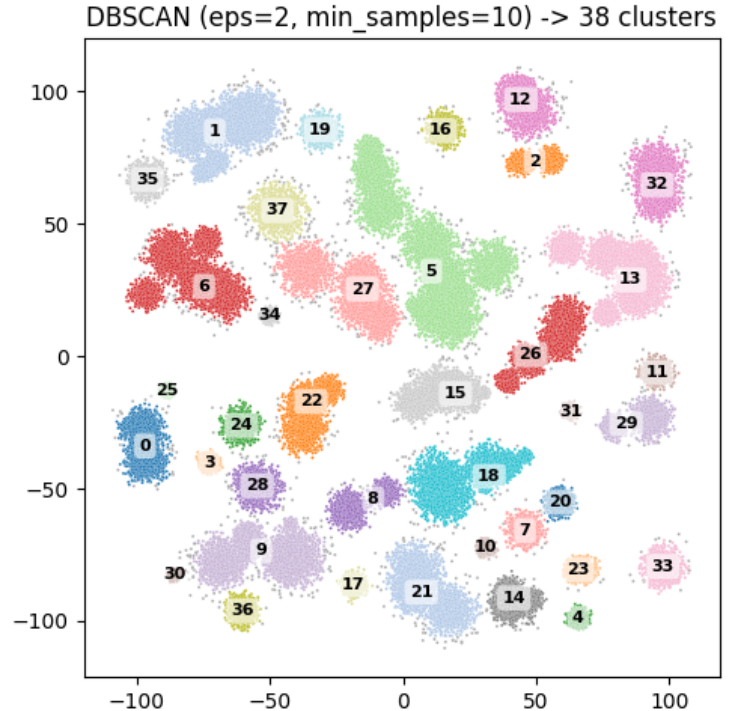


THE BIRCH CLASSIFICATION PROBLEM

DBSCAN

Chosen degrees of freedom:

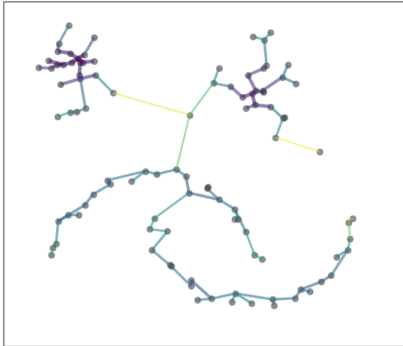
- ▶ Radius "eps" = 2.0
- ▶ Threshold number of neighbors = 10
- ▶ Metric: Euclidian distance



THE BIRCH CLASSIFICATION PROBLEM

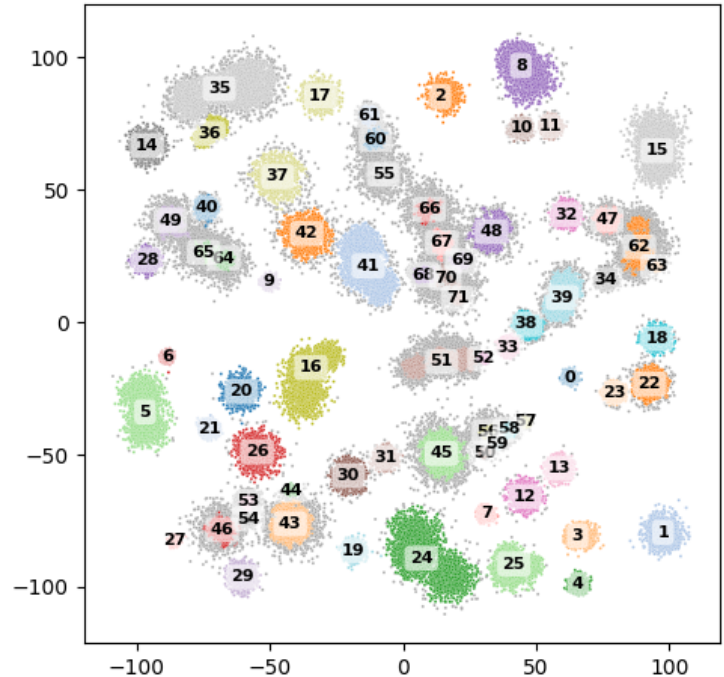
OTHER CLUSTERING: HDBSCAN

- Hierarchical DBSCAN builds density hierarchy
- Only min. points to form a cluster to be set, no "eps"



Source: https://hdbscan.readthedocs.io/en/latest/how_hdbscan_works.html, as of 02.03.2026

HDBSCAN (min_cluster_size=200) -> 72 clusters



THE BIRCH CLASSIFICATION PROBLEM

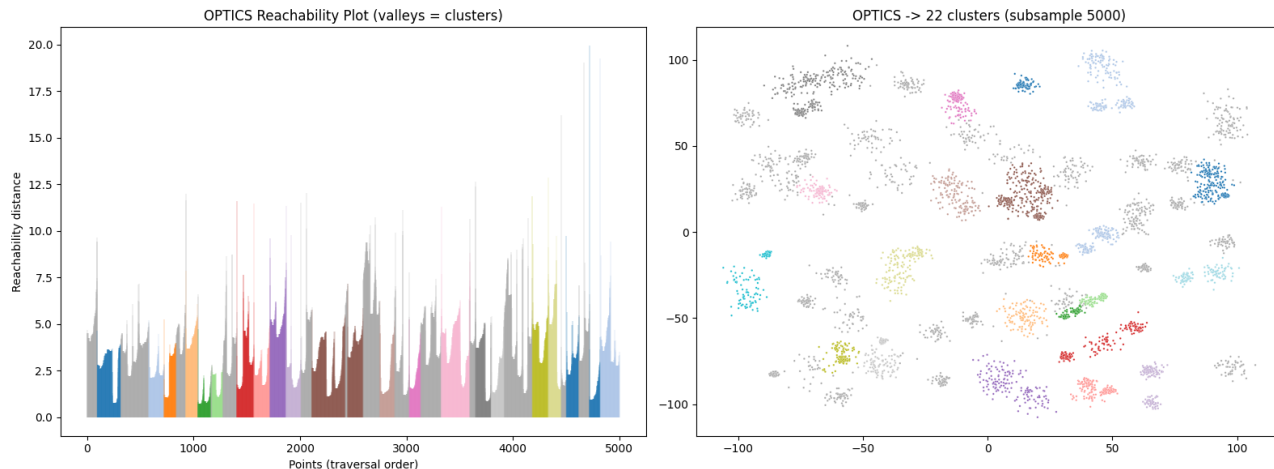
OTHER CLUSTERING: OPTICS

OPTICS — *Ordering Points To Identify Clustering Structure*

- ▶ Generalises DBSCAN: **no fixed** ϵ s needed
- ▶ Orders points so nearby points are neighbours; **valleys** in the reachability plot = clusters
- ▶ Handles clusters of **varying density** naturally
- ▶ $O(n \log n)$ — run on 5 000-pt subsample to limit runtime
- ▶ Found **22 clusters**, 33% noise (sparse subsample effect)

THE BIRCH CLASSIFICATION PROBLEM

OTHER CLUSTERING: OPTICS



OPTICS: reachability plot (left) and clustering result on 5 K subsample (right)

THE BIRCH CLASSIFICATION PROBLEM

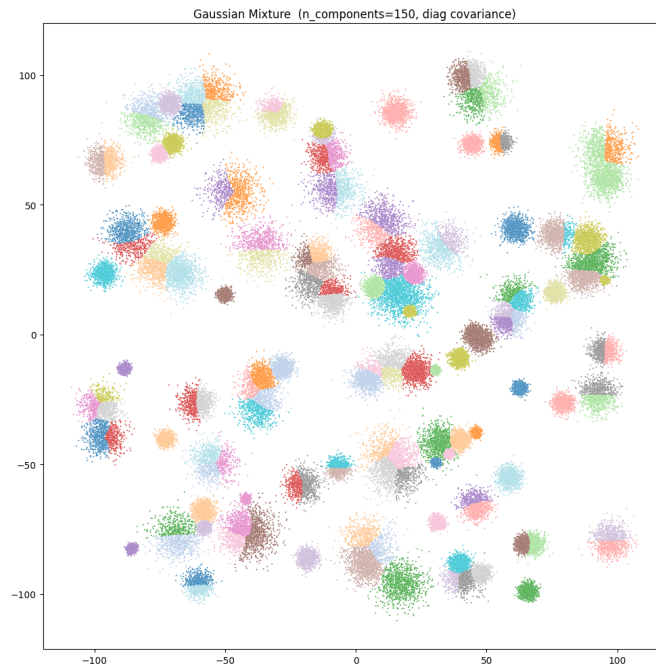
OTHER CLUSTERING: GMM

GMM — *Gaussian Mixture Model*

- ▶ Each cluster = one Gaussian component (**soft** probabilistic assignment)
- ▶ Unlike K-Means: elliptical shapes and **unequal cluster sizes** allowed
- ▶ Number of components selected via **BIC** (lower = better fit/complexity trade-off)
- ▶ BIC kept decreasing $\Rightarrow$ best $n = 150$ — over-splits the true clusters
- ▶ Weakest silhouette of all methods (0.46), high Davies-Bouldin (0.93)

THE BIRCH CLASSIFICATION PROBLEM

OTHER CLUSTERING: GMM



GMM ($n = 150$ components, diagonal covariance)

HOW DO WE MEASURE CLUSTERING QUALITY?

Metric	Formula	Meaning
Silhouette Score	$s(i) = \frac{b(i) - a(i)}{\max(a(i), b(i))}$	$a(i)$: mean intra-cluster distance. $b(i)$: mean distance to nearest other cluster. Range $[-1, 1]$; higher = more compact & well-separated.
Davies-Bouldin Index	$\frac{1}{K} \sum_{i=1}^K \max_{j \neq i} \frac{\sigma_i + \sigma_j}{d(c_i, c_j)}$	σ_i : avg distance of points in cluster i to its centroid. $d(c_i, c_j)$: distance between centroids. Range $[0, \infty)$; lower = better separated clusters.

METRICS COMPARISON

Complexity & Quality — BIRCH-3 ($N = 100\,000$ points)

Method	Complexity	# Clusters	Silhouette $\uparrow$	Davies-Bouldin $\downarrow$
K-Means ($K = 50$)	$O(nKl)$	50	0.568	0.585
DBSCAN	$O(n \log n)$	38	0.473	0.529
HDBSCAN	$O(n \log n)$	72	0.645	0.437
OPTICS (5K sub)	$O(n \log n)$	22	0.641	0.518
GMM ($n = 150$)	$O(nKl)$	150	0.465	0.929
BIRCH	$O(n)$	50	0.525	0.646

⇒ **HDBSCAN**: best silhouette (0.645) **and** best Davies-Bouldin (0.437)

BREAD BASKET — DATA STRUCTURE

Each row = **one item** in a transaction:

- ▶ **Transaction**: basket ID
- ▶ **Item**: product name
- ▶ **date_time**: timestamp
- ▶ **period_day**: morning / afternoon / evening
- ▶ **weekday_weekend**: day type

Tr.	Item	Date	Period
1	Bread	30-10-2016	morning
2	Scandinavian	30-10-2016	morning
3	Hot chocolate	30-10-2016	morning
3	Jam	30-10-2016	morning
3	Cookies	30-10-2016	morning
...	...	...	...
9682	Coffee	09-04-2017	afternoon
9682	Tea	09-04-2017	afternoon
9683	Pastry	09-04-2017	afternoon

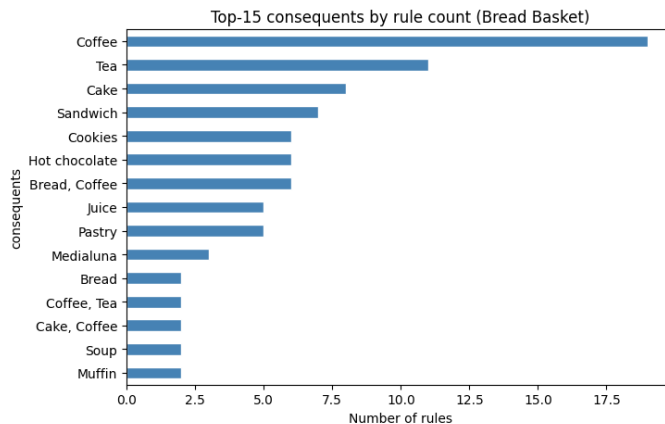
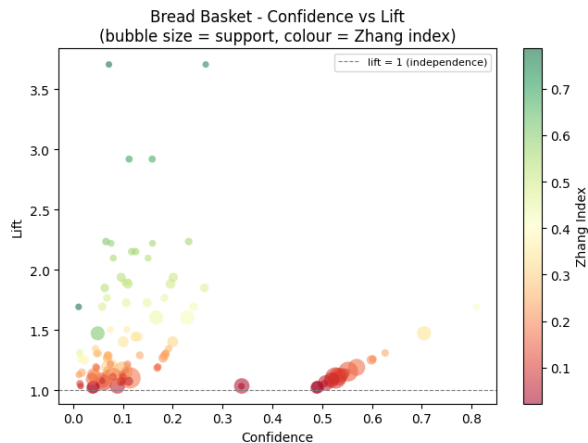
20 507 rows · 9 684 transactions · ~60 unique items

Top items: **Coffee** (47.8%), **Bread** (32.7%), **Tea** (14.3%)

ASSOCIATION RULE METRICS

Metric	Formula	Interpretation
Support	$P(A \cup B)$	Frequency of the pair. Low = rare rule.
Confidence	$\frac{P(A \cup B)}{P(A)}$	Given A is bought, how likely is B ? Directional.
Lift	$\frac{P(A \cup B)}{P(A)P(B)}$	>1 : positive assoc.; $=1$: independent; <1 : substitutes.
Leverage	$P(A \cup B) - P(A)P(B)$	Absolute co-occurrence surplus over independence.
Conviction	$\frac{1 - P(B)}{1 - P(B A)}$	How much the rule fails if items were independent.
Zhang	$\frac{P(AB) - P(A)P(B)}{\max[. . .]}$	Symmetric $\in [-1, 1]$; robust when support is low.

BREAD BASKET — KEY ASSOCIATION RULES



Confidence vs. Lift (bubble size = support, colour = Zhang index)

BREAD BASKET — STRONGEST RULES

By lift / Zhang:

Rule	Lift	Conf.	Zhang
Sandwich $\Rightarrow$ Coke	3.71	0.07	0.79
Coke $\Rightarrow$ Sandwich	3.71	0.27	0.74
Juice $\Rightarrow$ Cookies	2.92	0.16	0.68
Cookies $\Rightarrow$ Juice	2.92	0.11	0.70

By confidence:

Rule	Conf.	Lift
Keeping It Local $\Rightarrow$ Coffee	0.81	1.69
Toast $\Rightarrow$ Coffee	0.70	1.47
Salad $\Rightarrow$ Coffee	0.63	1.31

BREAD BASKET — SHOP RECOMMENDATIONS

- ▶ **Bundle Sandwich + Coke**: lift = 3.7, Zhang = 0.79 — strongest cross-category signal. A combo meal deal should increase basket size.
- ▶ **Co-locate Juice & Cookies**: lift = 2.9, symmetric Zhang ≈ 0.7 — customers buying either frequently buy both. Place on adjacent shelves.
- ▶ **Coffee anchors the menu** (48% support). Toast, Pastry, and Medialuna all have high confidence towards Coffee — group as a “morning set” near the coffee counter.
- ▶ **Cake & Tea** co-occur (lift ≈ 1.6 , Zhang ≈ 0.43): consider an afternoon tea-and-cake promotion on weekends.
- ▶ Rules with high Zhang but *low* support (Sandwich/Coke, Juice/Cookies) indicate **niche but reliable pairings** — ideal for targeted bundle pricing rather than broad promotions.

GROCERIES — DATA STRUCTURE

Each row = **one item purchased by one member**:

- ▶ **Member_number**: customer ID (basket key)
- ▶ **Date**: purchase date
- ▶ **itemDescription**: product name
- ▶ Also contains year / month / day / day_of_week

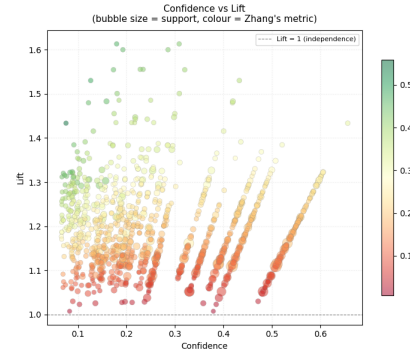
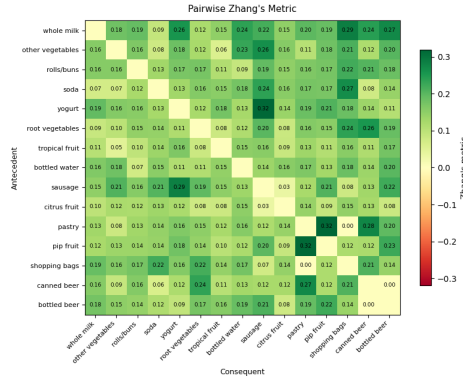
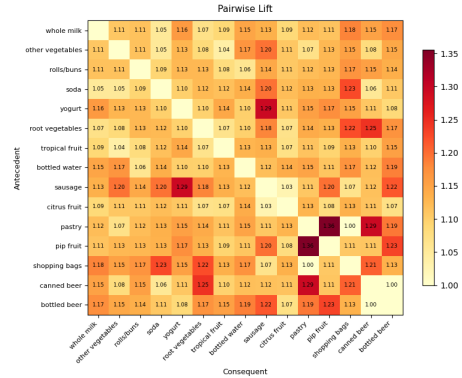
Member	Item	Date
1808	tropical fruit	2015-07-21
2552	whole milk	2015-05-01
2300	pip fruit	2015-09-19
1187	other vegetables	2015-12-12
3037	whole milk	2015-01-02
...	...	...
4471	sliced cheese	2014-08-10
1097	cake bar	2014-04-16
1510	fruit/veg juice	2014-03-12

38 765 rows · 3 898 members · 167 unique items

Top items: **whole milk** (45.8%), **other vegetables** (37.7%), **rolls/buns** (34.9%)

GROCERIES — KEY ASSOCIATION RULES

Groceries — Association Rule Metrics (top-15 items)



Confidence vs. Lift (bubble size = support, colour = Zhang index)

GROCERIES — STRONGEST RULES

By lift / Zhang:

Rule	Lift	Conf.	Zhang
$\{\text{veg, milk}\} \Rightarrow \{\text{buns, yogurt}\}$	1.61	0.18	0.47
$\{\text{buns, yogurt}\} \Rightarrow \{\text{veg, milk}\}$	1.61	0.31	0.43
$\{\text{buns, milk}\} \Rightarrow \{\text{veg, yogurt}\}$	1.60	0.19	0.46

By confidence:

Rule	Conf.	Lift
$\{\text{buns, veg, yogurt}\} \Rightarrow \text{whole milk}$	0.66	1.43
$\{\text{bottled water, yogurt}\} \Rightarrow \text{whole milk}$	0.61	1.32

whole milk appears in nearly every high-support rule — it is the *anchor* of this dataset.

GROCERIES — SHOP RECOMMENDATIONS

- ▶ **Cluster the dairy & fresh-produce aisle:** whole milk, other vegetables, rolls/buns, yogurt, and sausage form a dense co-purchase group — proximity reduces friction and boosts cross-sells.
- ▶ **Promote whole milk as a bundle anchor:** it appears in $>60\%$ of all significant rules (support up to 19%). A “meal-prep basket” coupon (milk + veg + buns) can lift average basket value.
- ▶ **Yogurt + rolls/buns + sausage** (lift ≈ 1.55): a breakfast combo — promote on end-caps or create a weekend breakfast deal.
- ▶ Lift values are lower here than in Bread Basket (max ≈ 1.6 vs. 3.7) because baskets aggregate *all* purchases per member. Focus on **category adjacency** over individual item pairing.
- ▶ High-confidence rules ($\{\text{buns}, \text{veg}, \text{yogurt}\} \Rightarrow \text{milk}$, conf = 0.66) suggest **targeted digital coupons** when a member adds 2–3 of these items to an online cart.

SUMMARY

- ▶ **Clustering:** HDBSCAN outperforms all methods on BIRCH-3 — best silhouette (0.645) and Davies-Bouldin (0.437); handles varying cluster sizes with no ϵ s to tune
- ▶ **Association (Bread Basket):** Sandwich/Coke (lift 3.7) and Juice/Cookies (lift 2.9) are the strongest pairs; Coffee anchors the menu
- ▶ **Association (Groceries):** whole milk is the universal anchor; dairy/produce cluster drives most high-support rules
- ▶ **Key takeaway:** high Zhang + high lift = actionable bundle/placement recommendation; high support alone may just reflect popular standalone items

THE END

Thank you for your attention!

Questions?